Karen Deng

0490004948 | keyuedeng6@gmail.com | https://karendeng.com/

SUMMARY

Computing and Software Systems student at the University of Melbourne with hands on experience building production-grade systems. Currently a Systems Engineer Intern at Cisco, delivering AI-integrated tools, data platforms, and automation solutions for enterprise clients. Passionate about using data, AI, and modern engineering practices to build systems that meaningfully change how people and organisations operate.

EDUCATION

Bachelor of Science in Computing and Software Systems Mar, 2024 - Oct, 2027
The University of Melbourne
- WAM: 80 (First Class Honours)
- Relevant coursework: Algorithms & Data Structures, Database Systems, Object Oriented Programming

EXPERIENCE

Systems Engineer Intern Mar, 2026 - Sep, 2026
Cisco, Melbourne, Australia
- Built a production data ingestion pipeline (Splunk Add-on) for Rio Tinto, implementing OAuth2 and JWT lifecycle management to securely stream real-time network and fleet health data from Cisco's REST API into an enterprise analytics platform — enabling large-scale operational visibility across thousands of devices
- Developed a Python automation tool that discovers and maps live network topologies by querying network devices via SSH, transforming raw CLI data into exportable visual diagrams; reducing manual documentation effort for network operations teams
- Redesigned Cisco's AI-driven network operations workshop (ThousandEyes, Meraki, Splunk), tested MCP integrations end-to-end, and delivered an interactive HTML lab guide for enterprise attendees — translating complex technical concepts into accessible, customer-facing learning experiences

RELEVANT PROJECT WORK

NLP-Driven Journal Knowledge Graph (moire-xi.vercel.app) Dec, 2025 - Jan, 2026
Next.js, React, Prisma ORM, PostgreSQL, Clerk, OpenAI
- Designed and deployed a full-stack AI application that uses vector embeddings and NLP to extract topics, analyse sentiment, and generate LLM-powered summaries from unstructured text — processing raw data into structured, queryable insights stored in PostgreSQL
- Built an interactive D3 force-directed visualisation to surface semantic relationships between topics, making complex data immediately understandable to end users

AI Medical Case Simulator (devpost.com/software/medcase) Jan, 2026 - Jan, 2026
React, TypeScript, Node.js, Express, FastAPI, OpenAI
- Architected a full-stack AI simulation platform at a 36-hour hackathon, integrating OpenAI's LLM with a FastAPI microservice and real-time voice synthesis to deliver low-latency, voice-enabled clinical conversations

OTHER EXPERIENCE

Python Developer
DSCubed Winter Program - University of Melbourne, Melbourne, Australia
- Built an AI-powered modular CLI assistant integrating OpenAI with async function calling, reducing new member onboarding from 1+ hours to 15 minutes by automating account creation and resource allocation via n8n workflows

Project Consultant Feb, 2025 - Jul, 2025
180 Degrees Consulting - University of Melbourne, Melbourne, Australia
- Delivered a 12-week project for Creative Australia, co-authoring a 30+ page impact evaluation framework with KPIs, methodologies, and data collection strategies to assess climate initiatives in the arts sector

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, C, SQL
Frameworks/Libraries: React, Next.js, Node.js, Express, FastAPI, Prisma ORM, D3.js, NetworkX, TextFSM
Technologies: PostgreSQL, OpenAI API, MCP, Splunk (UCC Framework), REST/JSON APIs, OAuth2, Git, Clerk
Certifications: Cisco CCNA (in progress), Cisco AI Technical Practitioner